

Henry Yu

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EDUCATION

Stony Brook University

Stony Brook, New York

Bachelor of Science in Computer Science, Focus in Algorithms and Systems

May 2027

Double major in Applied Mathematics and Statistics, Concentration in Probability and Optimization

Relevant Coursework: Data Structures, Machine Learning, Algorithms, Analysis of Algorithms, Graph Theory, Systems Fundamentals, Computer Networks, Theory of Computation, Linear Algebra

PROJECTS

PathPulse – eBPF TCP Observability Agent | C, Go, eBPF (CO-RE), Linux

August 2026-Present

- Wrote **4 eBPF** kernel hooks (fentry, fexit, tp_btf) in **C** on TCP connect, state-transition, and retransmit paths, capturing connection outcomes without application instrumentation.
- Built a lock-free atomic state machine classifier guaranteeing exactly one outcome per tracked connection, validated by exhaustive permutation testing over every reachable hook ordering.
- Diagnosed a null-address bug caused by kernel timing (fentry firing before local address assignment), fixed by splitting address capture across fentry and fexit hooks.

SEC Filing Evaluation Harness | Python, RAG, Anthropic Claude API, pandas, pytest

August 2026

- Built an evaluation harness for numerical accuracy on SEC filings against the open **LEDGER** benchmark, backed by an **80+** test suite in **pytest**.
- Generated **3,700+** questions on computed metrics (margins, ratios, YoY growth) from **SEC EDGAR XBRL** ground truth, extending the benchmark past extraction alone.
- Ran a diagnostic evaluation over **25** questions across **~20** companies: **72%** of answers found the right figure at the wrong scale, masked by a **4%** raw accuracy score.

Market Data Engine | C++, MySQL, GoogleTest, NASDAQ ITCH 5.0

May 2026 – August 2026

- Wrote a **C++** parser for the **NASDAQ ITCH 5.0** binary protocol, reconstructing order books across **5** symbols from **55,748** raw messages at **~550 MB/sec** throughput, verified by **61 GoogleTest** cases.
- Built a **MySQL** persistence layer with batch ingestion and snapshot capture, plus analytics queries (VWAP, rolling volatility, spread/depth, order-to-trade ratio) over real NASDAQ tick data.

WORK EXPERIENCE

BetterTherapy

New York, NY

Software Engineering Intern

May 2025 – August 2025

- Designed a **SQLAlchemy** data layer over **SQLite** for an automated LLM evaluation pipeline: schema with a reusable timestamp mixin, cascading request/response relationships, and a session decorator.
- Wrote atomic upserts (ON CONFLICT DO UPDATE) for a node blacklist, incrementing violation counts safely under concurrent writes.
- Avoided N+1 round trips with **selectinload** eager loading, added bulk inserts, and polled the **OpenAI Batch API** on 24-hour windows.

ReWild Long Island

Long Island, NY

Software Engineer, Dev Team

August 2026 - Present

- Architected and containerized a full-stack backend (**FastAPI**, **PostgreSQL 16**, pgvector) with **Docker Compose** and **Alembic** migrations; designed an 11-table schema with composite keys, custom enums, and a many-to-many crosswalk for 100+ Long Island communities.

Stony Brook University

Stony Brook, New York

Undergraduate Teaching Assistant

August 2026 - Present

CSE 260: Honors Data Structures & Programming Abstractions

- Run lab sections for **~80** students with a team of 4 TAs, guiding implementation and analysis of data structures and programming abstractions.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Go, SQL, R, Java, C/C++, Bash, HTML/CSS

Frameworks/Tools: React, Node.js, Express, Django, MongoDB, PostgreSQL, MySQL, SQLAlchemy, Git/GitHub, REST APIs, JWT Authentication, Linux, eBPF, AWS, Docker, Kubernetes, GoogleTest, PyTorch/TensorFlow

Libraries: pandas, NumPy, statsmodels, matplotlib

Concepts: Data Structures & Algorithms, OOP, Full-Stack Web Development, System Design, API Design, Event-Driven Architecture, Concurrency, State Management, Testing (pytest), CI/CD